

# Chemistry with Quantum Computer Simulators

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Division/Area: Coding, Experimental Research

Program : Experiences in Research

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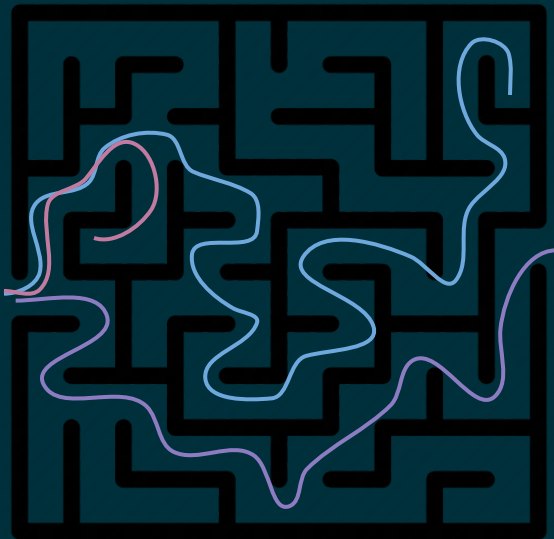
# What is Quantum Computing?

## Classical Properties

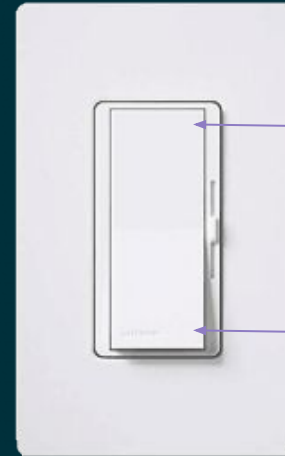


On (1 state)

Off (0 state)

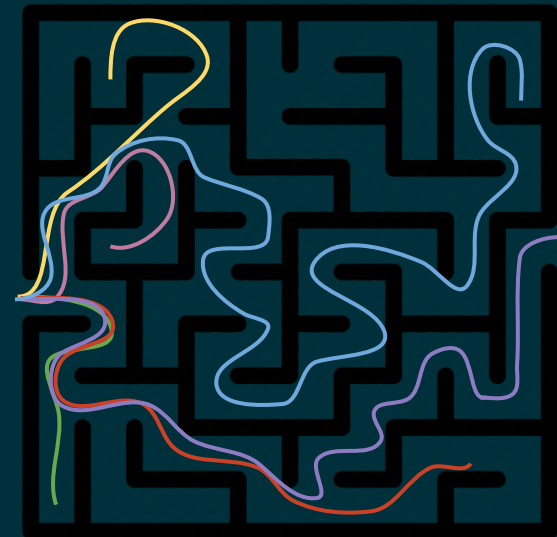


## Quantum Properties



On (1 state)

Off (0 state)



# What is a Variational Quantum Eigensolver?

## Quantum Computer

Takes in the ansatz or wavefunction representation of the molecule and calculates the energy.



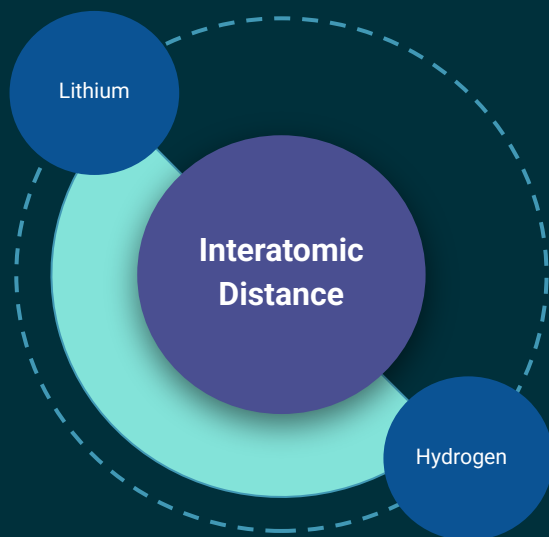
## Classical Computer

Varies the ansatz to try to get a lower energy value

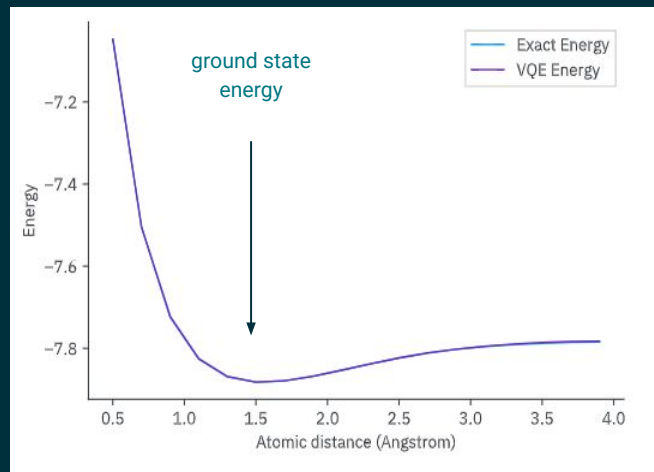


## Variational Quantum Eigensolver

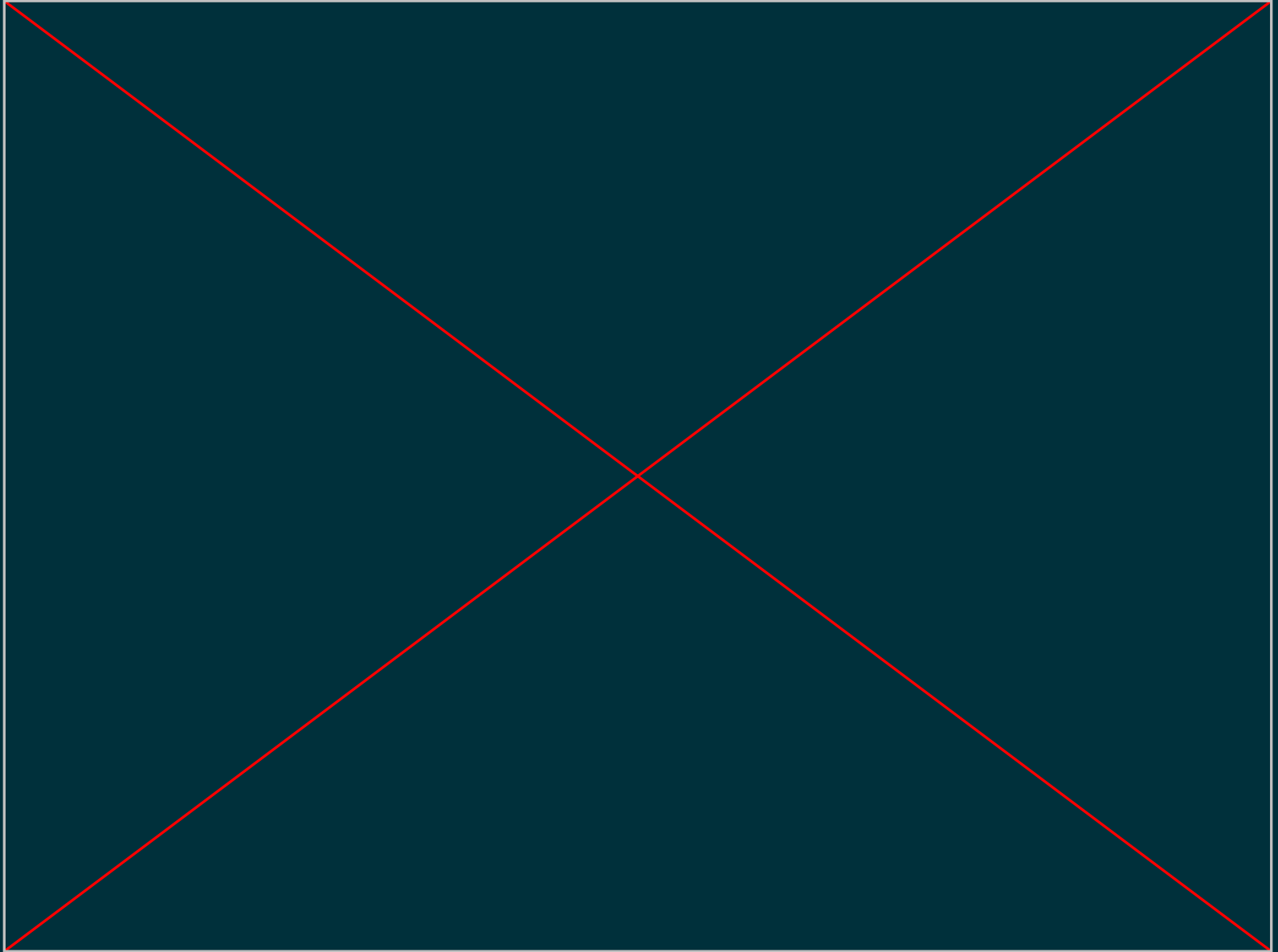
Quantum and Classical parts work together to calculate the ground state energy of the molecule at different interatomic distances



*LiH potential energy curve*



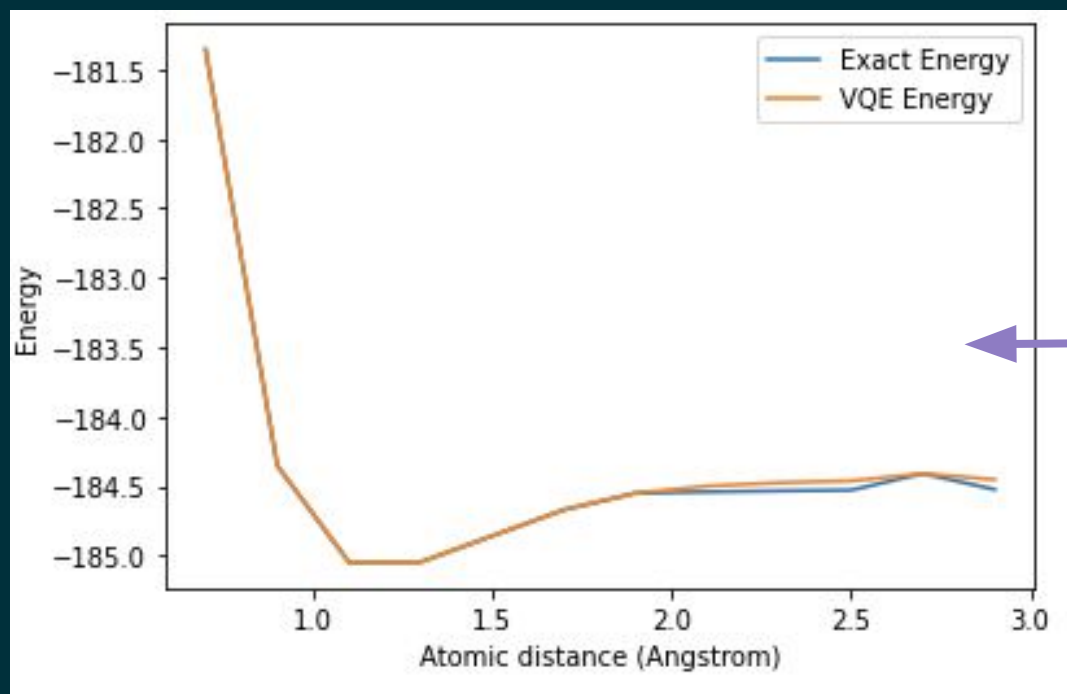
# What are Chemistry Simulations?



# What are Chemistry Simulations?

|                           |                        |                          |
|---------------------------|------------------------|--------------------------|
| Interatomic Distance: 0.7 | VQE Result: -181.35603 | Exact Energy: -181.35603 |
| Interatomic Distance: 0.9 | VQE Result: -184.35277 | Exact Energy: -184.35277 |
| Interatomic Distance: 1.1 | VQE Result: -185.04872 | Exact Energy: -185.04872 |
| Interatomic Distance: 1.3 | VQE Result: -185.04712 | Exact Energy: -185.04712 |
| Interatomic Distance: 1.5 | VQE Result: -184.85825 | Exact Energy: -184.85825 |
| Interatomic Distance: 1.7 | VQE Result: -184.66998 | Exact Energy: -184.66998 |
| Interatomic Distance: 1.9 | VQE Result: -184.55050 | Exact Energy: -184.55050 |
| Interatomic Distance: 2.1 | VQE Result: -184.50014 | Exact Energy: -184.54147 |
| Interatomic Distance: 2.3 | VQE Result: -184.47550 | Exact Energy: -184.53430 |
| Interatomic Distance: 2.5 | VQE Result: -184.46280 | Exact Energy: -184.52915 |
| Interatomic Distance: 2.7 | VQE Result: -184.41010 | Exact Energy: -184.41010 |
| Interatomic Distance: 2.9 | VQE Result: -184.45299 | Exact Energy: -184.52433 |

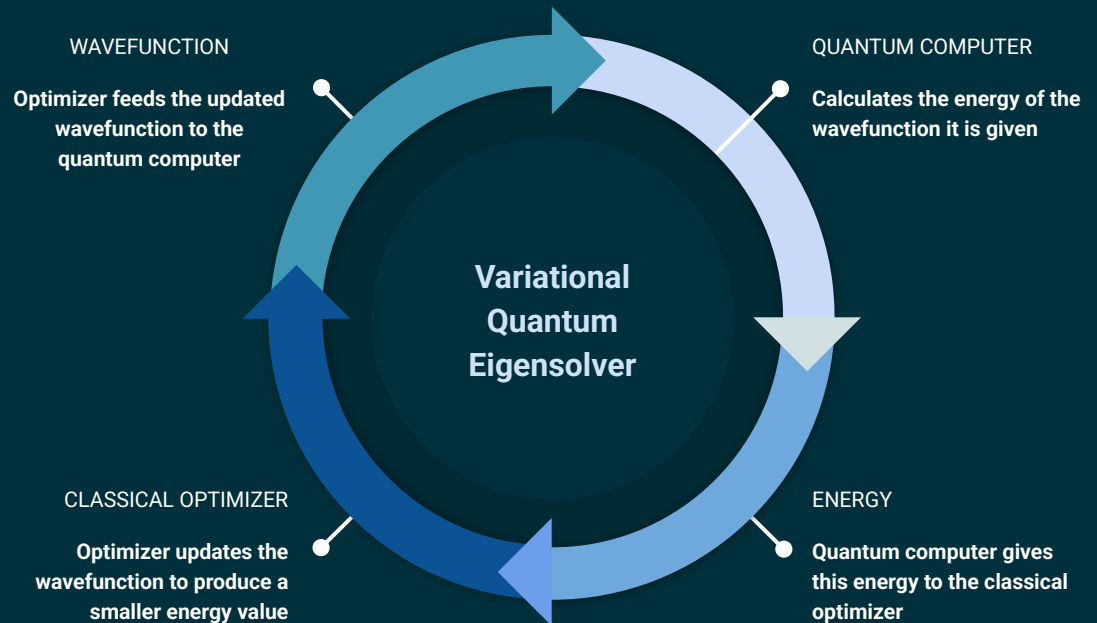
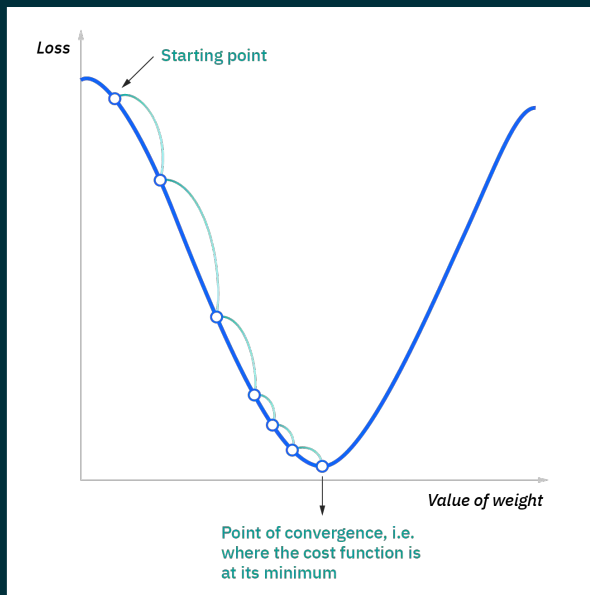
Numerical  
Representation



Graphical  
Representation

# Optimization of the VQE

- VQE goes through the process of optimization for each interatomic distance
- Optimizer is trying to reach the bottom of the valley- the minimum energy value



# Optimization of the VQE

## OPTIMIZERS

Tried out different optimizers

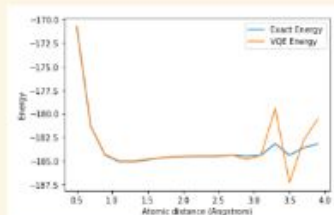
COBYLA worked best & SLSQP worked second best as they were constrained optimizers

## ITERATIONS

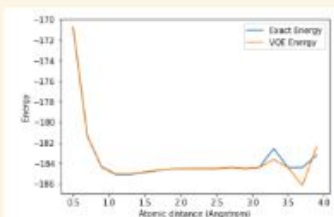
Experimented with different number of iterations

With more iterations, more refined, less likely to get stuck in local minimas

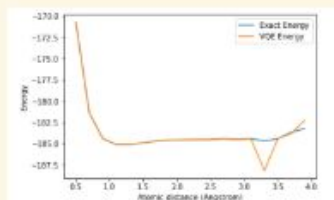
### CO<sub>2</sub> curve optimization



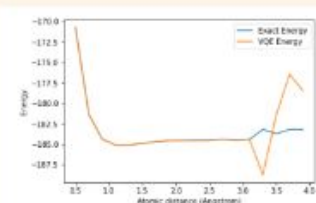
COBYLA, 10 iterations



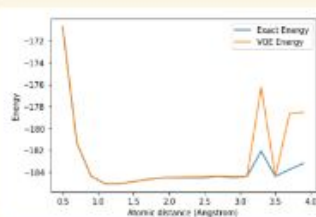
COBYLA, 50 iterations



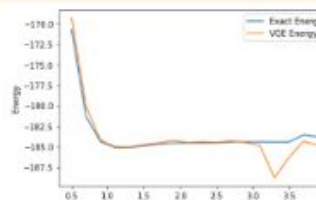
COBYLA, 200 iterations



SLSQP, 10 iterations



SLSQP, 50 iterations



SPSA, 50 iterations

# Using Different Backends

## ibmq\_armonk

System status ● Online  
Processor type Canary r1.2

Qubit QV  
**1 1**



## ibmq\_belem

System status ● Online - Queue paused  
maintenance

Processor type Falcon r4T

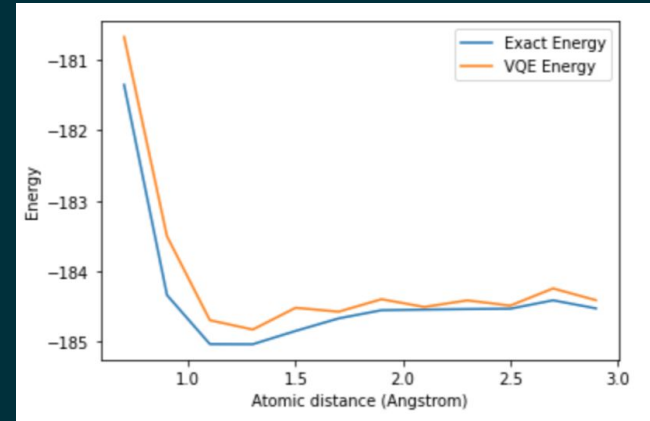
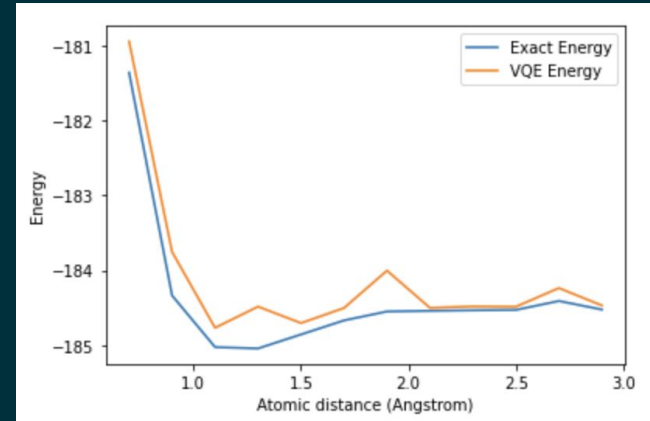
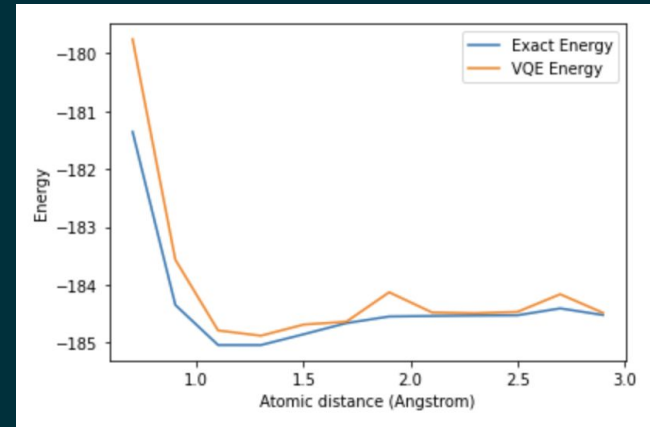
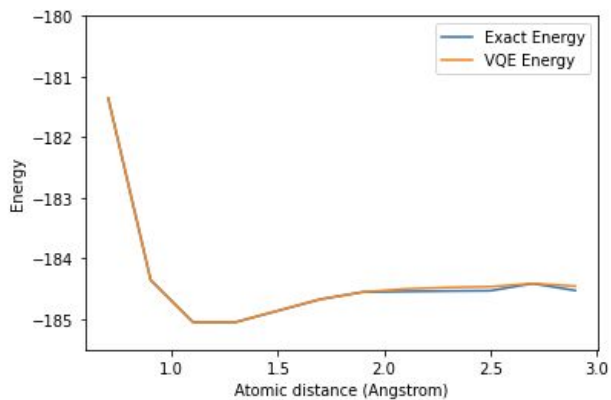
Qubits QV CLOPS  
**5 16 2.5K**



## ibmq\_nairobi

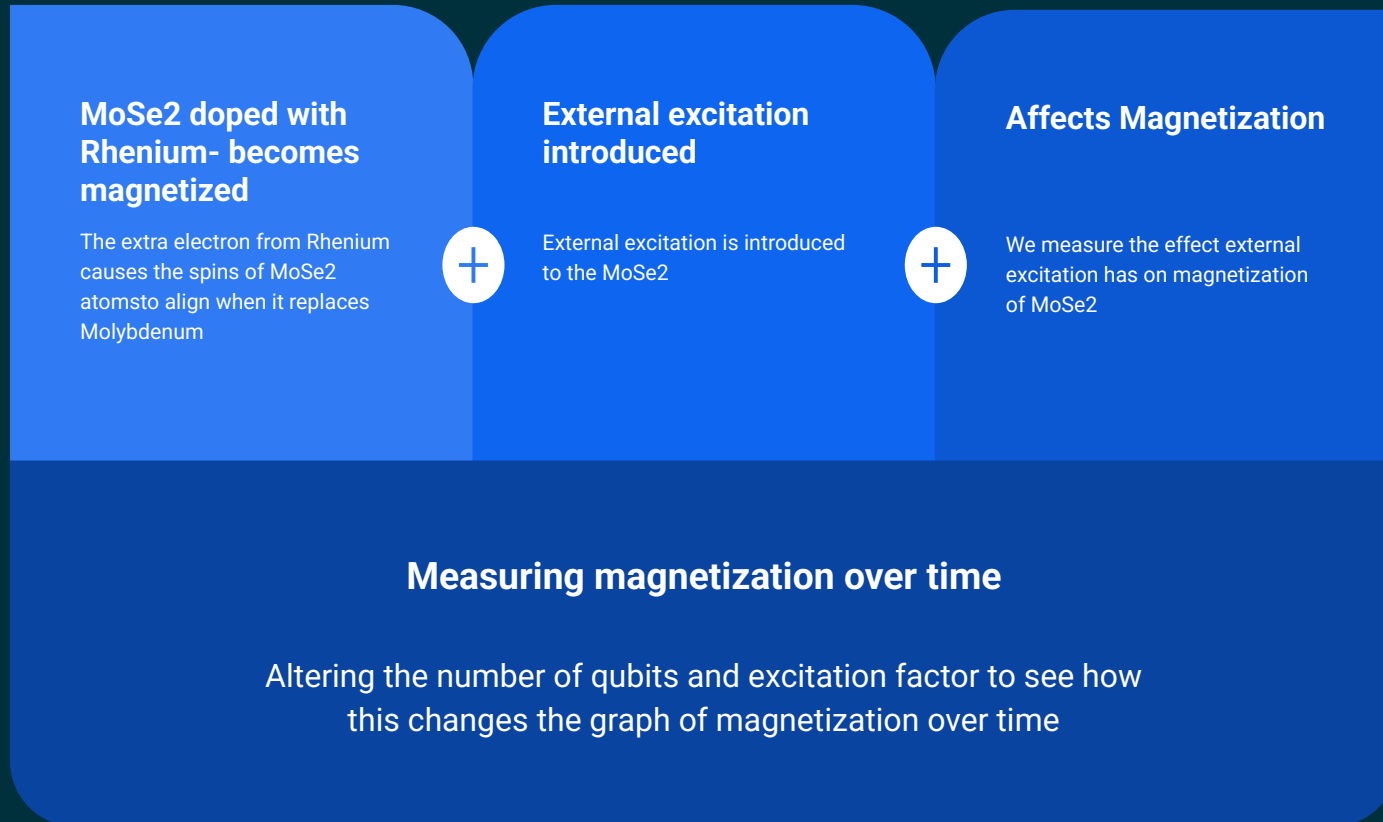
System status ● Online  
Processor type Falcon r5.11H

Qubits QV CLOPS  
**7 32 2.6K**



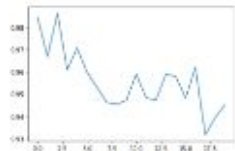


# Magnetization of MoSe2 over time

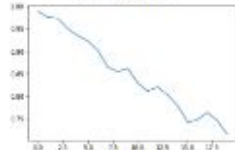


# Impact of different excitation factors across 2-5 atom chains

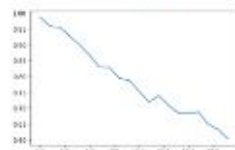
*Jz factor = 0.2*



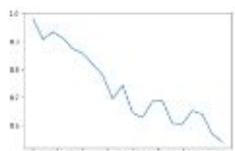
2 qubits



3 qubits

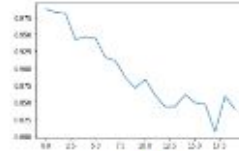


4 qubits



5 qubits

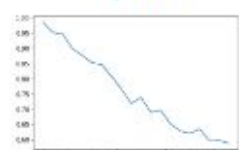
*Jz factor = 0.5*



2 qubits



3 qubits

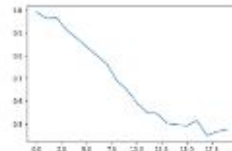


4 qubits

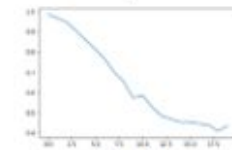


5 qubits

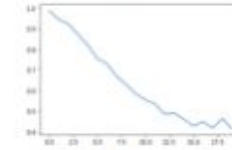
*Jz factor = 1.0*



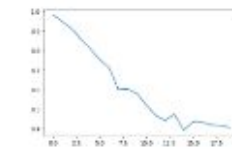
2 qubits



3 qubits

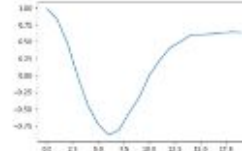


4 qubits

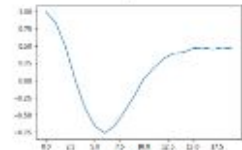


5 qubits

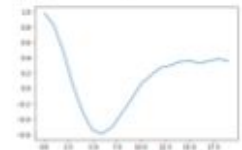
*Jz factor = 5.0*



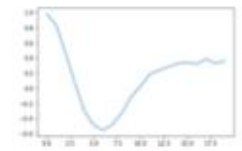
2 qubits



3 qubits

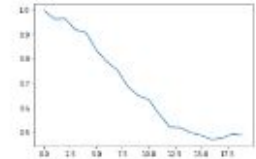


4 qubits

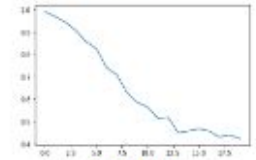


5 qubits

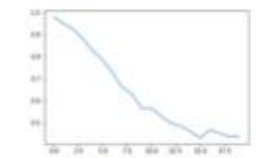
*Jz factor = -1.0*



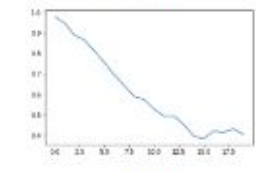
2 qubits



3 qubits

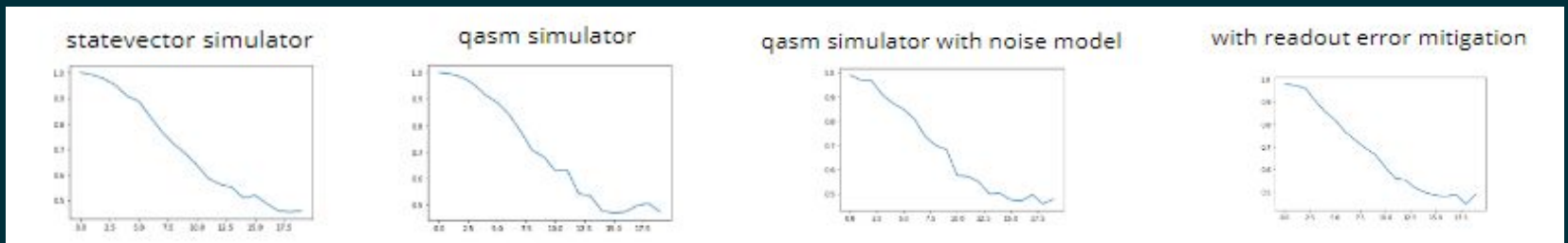
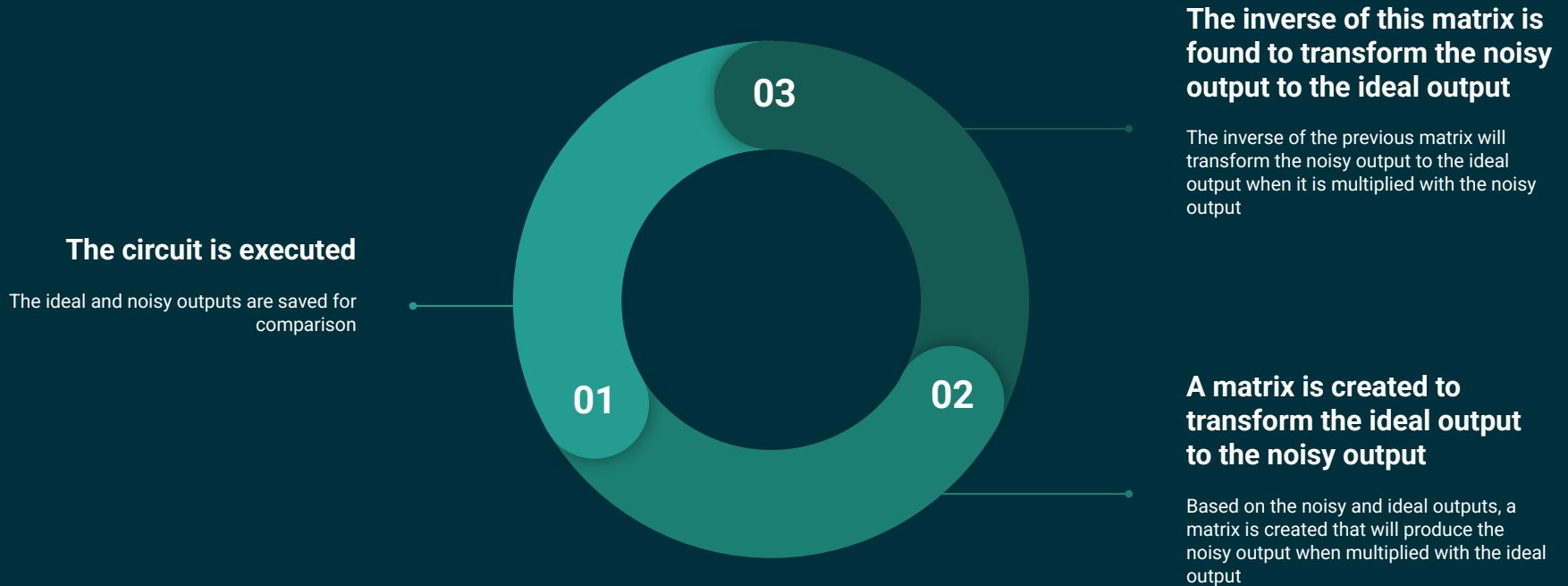


4 qubits

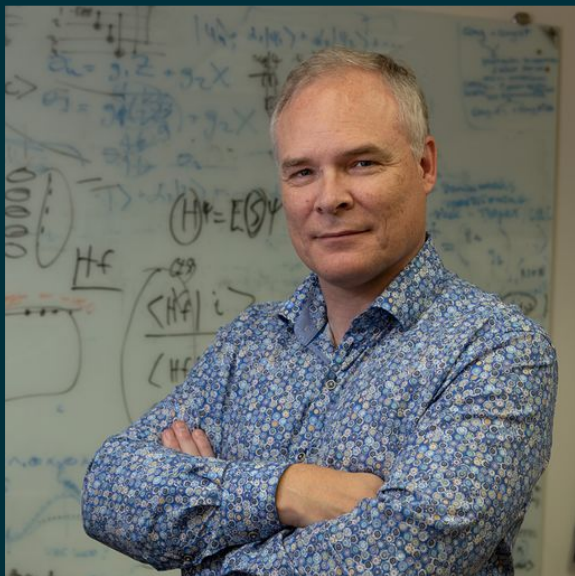


5 qubits

# Readout Error Mitigation



## Conclusion



Dr. Bert de Jong



Dr. Niladri Gomes



Dr. Lindsay Bassman



Qiskit



IBM Q

# Thank You

Questions?